

NENJIANG, China

WMO: 505570

Lat: 49.1561N

Lon: 125.2278E

Elev: 243

StdP: 98.44

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.0	-32.6	-38.1	0.1	-34.8	-35.9	0.1	-32.4	8.1	-17.3	7.3	-16.7	0.7	70	0.619

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	31.2	19.6	29.3	19.4	27.7	19.3	23.0	26.7	22.0	25.9	21.1	25.4	5.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.9	17.1	24.8	20.8	16.0	24.1	19.7	14.9	23.1	69.9	26.8	66.1	25.7	62.7	25.4	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.4	8.7	7.5	-36.5	34.7	3.3	2.0	-38.9	36.1	-40.8	37.3	-42.7	38.5	-45.1	39.9		
			-36.6	24.7	3.3	1.2	-39.0	25.6	-40.9	26.3	-42.8	27.0	-45.2	27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.3	-22.6	-17.9	-7.5	4.3	12.8	19.5	21.7	19.1	12.6	3.2	-10.2	-20.5
	DBStd	16.45	5.14	5.29	6.75	4.73	4.42	3.61	2.60	2.89	4.10	4.84	6.90	4.86
	HDD10.0	4324	1012	780	543	179	19	0	0	0	20	217	606	947
	HDD18.3	6448	1270	1013	802	420	179	28	4	24	176	471	856	1206
	CDD10.0	1152	0	0	0	10	108	284	362	284	98	5	0	0
	CDD18.3	233	0	0	0	0	9	62	108	49	4	0	0	0
	CDH23.3	2069	0	0	0	7	147	663	821	378	53	1	0	0
	CDH26.7	589	0	0	0	1	39	234	236	70	8	0	0	0
Wind	WSAvg	3.0	1.8	2.2	3.3	4.2	4.4	3.4	2.9	2.8	3.1	3.3	2.7	2.0
Precipitation	PrecAvg	481	3	3	5	21	35	72	141	112	60	17	6	5
	PrecMax	708	9	16	14	78	74	156	389	264	109	49	20	14
	PrecMin	339	0	0	0	0	2	18	31	18	8	0	0	1
	PrecStd	90	2	3	4	18	19	35	67	62	31	13	5	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-7.7	1.0	11.5	24.0	30.6	34.7	33.5	30.6	27.7	20.2	8.0	-5.3
		MCWB	-9.2	-2.4	3.8	11.0	15.1	19.5	21.3	20.9	18.1	11.6	3.4	-6.8
	2%	DB	-10.6	-3.1	7.9	19.6	27.4	31.4	31.2	28.5	24.7	16.3	4.5	-8.6
		MCWB	-11.9	-5.2	2.1	8.5	14.1	18.5	21.4	21.2	16.5	9.2	0.6	-9.8
	5%	DB	-12.7	-6.3	5.0	16.5	24.4	29.4	29.1	27.0	22.6	13.6	1.5	-11.0
		MCWB	-13.8	-8.0	0.3	7.1	12.7	18.0	20.9	20.4	15.3	7.3	-1.3	-12.1
	10%	DB	-14.7	-8.9	2.6	13.5	21.7	27.2	27.5	25.5	20.5	11.1	-0.8	-13.0
		MCWB	-15.5	-10.2	-1.0	5.6	11.6	17.2	20.6	19.7	14.3	6.0	-2.9	-13.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-9.2	-2.5	5.2	12.0	16.6	21.9	24.7	24.0	20.3	13.0	3.6	-6.6
		MCDB	-7.8	0.5	10.3	21.7	24.9	29.2	29.0	27.3	23.8	18.8	7.5	-5.3
	2%	WB	-11.8	-5.2	2.4	9.4	15.1	20.6	23.4	22.7	18.3	10.0	0.8	-9.8
		MCDB	-10.6	-3.1	7.1	17.8	24.2	27.9	27.4	25.8	22.3	14.6	4.1	-8.6
	5%	WB	-13.8	-8.0	0.7	7.8	14.0	19.5	22.6	21.7	16.7	8.1	-1.0	-12.0
		MCDB	-12.8	-6.4	4.6	15.4	22.3	25.6	26.5	25.2	20.6	12.3	1.2	-11.1
	10%	WB	-15.5	-10.2	-0.8	6.2	12.7	18.5	21.7	20.7	15.3	6.4	-2.7	-13.9
		MCDB	-14.7	-9.0	2.5	12.8	19.7	24.5	25.6	24.3	19.4	10.6	-0.8	-13.0
Mean Daily Temperature Range	5% DB	MDBR	11.0	13.3	12.4	12.5	12.6	11.3	9.7	10.5	12.4	11.0	10.0	10.0
		MCDBR	11.8	14.3	13.5	17.4	16.6	15.0	12.7	12.9	15.2	14.9	11.6	11.1
		MCWBR	10.8	12.4	9.0	9.6	7.4	5.8	5.2	6.4	7.8	8.9	8.6	10.2
	5% WB	MCDBR	11.7	14.3	12.6	15.4	13.7	11.8	9.4	10.0	11.9	13.0	11.0	11.0
		MCWBR	10.6	12.4	8.5	8.8	6.6	5.1	5.1	5.7	6.8	8.5	8.5	10.0
Clear-Sky Solar Irradiance	taub	0.276	0.301	0.356	0.455	0.529	0.514	0.491	0.440	0.394	0.371	0.309	0.284	
	taud	2.197	2.135	2.059	1.941	1.818	1.967	2.074	2.191	2.253	2.191	2.245	2.171	
	Ebn at Noon	767	832	838	786	743	760	772	796	797	736	715	691	
	Edh at Noon	77	105	136	171	202	175	155	132	112	100	74	69	
All-Sky Solar Radiation	RadAvg	1.82	3.03	4.45	5.11	5.64	5.95	5.46	4.95	4.07	2.81	1.73	1.36	
	RadStd	0.08	0.14	0.27	0.41	0.42	0.66	0.53	0.41	0.32	0.16	0.13	0.11	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page